



Data structure → way to store the data in an effective way

① Series → rows & a single column/feature

② DataFrame → rows & columns
too many columns/

Series

A Series is a one-dimensional array-like structure in Pandas that can hold a sequence of values of any data type (integers, floats, strings, Python objects, etc.).

feature (>1)

DataFrame

A DataFrame is a two-dimensional labelled data structure like a table of MySQL. It contains rows and columns, and therefore has both a row and column index. Each column can have a different type of value such as numeric, string, boolean, etc., as in tables of a database.

Pandas Series

Numpy arrays

①

we can define our own
labeled index to access
elements of an array

accessed by their
integer position
using numbers only

②

series require more
memory

Numpy array lesser
memory

③

If two series are not
aligned, NaN or
missing values are
generated

There is no concept
of NaN value in
the case of

52 Numpy arrays

Not
a
Number
NaN → missing
value/
error

a ——— 10
b ——— 20
c ——— 30

b ——— 5
c ——— 15
d ——— 25

a ——— NaN
b ——— 25
c ——— 45
d ——— NaN